a) Nutritional benefits

The composition of human milk is best suited to the infants and provide nutrients in easily digestible and bioavailable form.

i. Carbohydrate

Human milk is the sweetest milk due to the high amount of lactose. Lactose which is present in higher levels in human milk facilitates the absorption of magnesium and calcium and favours amino acid absorption and nitrogen retention.

ii. Protein

Human milk contains 1.1 g percent protein. Human milk has 20% casein and 80 % whey proteins which constitutes lactalbumin and lactoferrin. Lactalbumin has an amino acid pattern that mainly approaches that of body protein and provides more essential amino acids than casein. Breast milk contains amino acids specific for brain development. Human milk offers a high tryptophan to neutral amino acids ratio which controls brain serotonin synthesis. Human milk also contains binding proteins of thyroxine, corticosterol, vitamin D, folate and B12.

iii. Lipids

Lipids Present in human milk are unsaturated fat, essential fatty acids, prostaglandin precursors, fat soluble vitamins, steroids, phospholipids and cholesterol. Lipids involved in the development of brain are mostly long chain polyunsaturated fatty acids. These are abundant in the breast milk. Human milk contains cholesterol and is essential for synthesis of myelin of the nervous system. Presence of chlorine, acetylcholine, phospholipid precursors and carnitine ensure optimum metabolism and brain development.

iv. Minerals

Unlike vitamins, mineral content of human milk is minimally influenced by mother's stores and immediate intake of calcium, magnesium, phosphorus, iron, copper, zinc, sodium and potassium. Minerals in the breast milk are largely protein bound and balance to enhance bioavailability. Calcium content in human milk is 28 mg and calcium phosphorus ratio of 2:1 in human milk is favourable. Breast milk ensures better oxygen saturation and increases the bioavailability of trace elements like copper, cobalt, selenium, iron and zinc.

v. Vitamins

Breast milk contains more vitamin A, C and E than cow's milk. Breast milk contains water soluble vitamin D along with fat soluble fraction which protects against rickets. riboflavin, pyridoxine and B12 content of human milk are also related to the dietary intake of mother. Heat liable vitamins like thiamine and ascorbic acid are completely available in human milk. Breast fed infants receive about 25 -30µg of folate daily, most of which is available for absorption.

b) Hormones and growth factor benefits

Breast milk is so rich source of hormones like thyroid stimulating hormone (TSH), thyroxin, parathyroid hormone, corticosteroids, calcitonin, erythropoietin, oxytocin, growth hormone releasing factor, insulin and prolactin. It contains growth regulating factors, growth promoting factors and growth modulators.

c) Immunological benefits

These factors are present in colostrum as well as in matured milk.

i. Macrophages

Human milk contains macrophages. They contribute immunity in two ways: (a) They engulf and digest bacteria. (b) These cells synthesize complement, a protein involved in establishing immunity to infectious organism.

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